ClipSAT : IB Diploma Mathematics: Analysis & Approaches / Applications [ANSWER KEY]

Unit 2.01: Topic 2: Functions, Transformations & Modeling

Q1. The graph of the function fx = -x^2 + 4x - 3 is shown below. What are the coordinates of the y-intercept?

[x] A: 0, -3
[ ] B: 0, 3
[ ] C: -1, 0
[ ] D: 3, 0

Solution: The y-intercept of a function occurs when x=0 . Substituting x=0 into the function fx = -x^2 + 4x - 3 gives f0 = -0^2 + 40 - 3 = -3 . Therefore, the coordinates of the y-intercept are 0, -3 . [M1] [A1]

Q2. Consider the function fx = 2x + 3 . What is the value of f4 ?

[ ] A: 7
[x] B: 11
[ ] C: 10
[ ] D: 5

Solution: To find the value of f4 , we substitute x=4 into the function fx = 2x + 3 . f4 = 24 + 3 f4 = 8 + 3 f4 = 11 [M1] [A1]

Q3. Consider the function fx = 2x - 3 . Find the value of f3 .

[x] A: 3
[ ] B: 6
[ ] C: 9
[ ] D: 0

Solution: To find the value of f3 , we substitute x=3 into the function definition fx = 2x - 3 . f3 = 23 - 3 f3 = 6 - 3 f3 = 3 . [M1] [A1]

Q4. Consider the function fx = -x^2 + 6x - 5 . Determine the coordinates of the vertex of the graph of y = fx .

[x] A: 3, 4
[ ] B: -3, 4
[ ] C: 3, -4
[ ] D: 4, 3

Solution: The given function is a quadratic of the form fx = ax^2 + bx + c , where a = -1 , b = 6 , and c = -5 . The x-coordinate of the vertex of a parabola is given by x = -frac{b}{2a} . Substituting the values, we get x = -frac{6}{2-1} = -frac{6}{-2} = 3 . To find the y-coordinate of the vertex, substitute x=3 into the function: f3 = -3^2 + 63 - 5 = -9 + 18 - 5 = 9 - 5 = 4 . Therefore, the coordinates of the vertex are 3, 4. [M1] [A1]

Q5. Consider the function fx = 2x - 1 . Find the value of f3 .

[x] A: 5
[ ] B: 6
[ ] C: 7
[ ] D: 4

Solution: fx = 2x - 1 . To find f3 , substitute x=3 into the function: f3 = 23 - 1 . f3 = 6 - 1 . f3 = 5 . Therefore, the value of f3 is 5. [M1] [A1]

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Q6. Consider the function fx = x^2 - 4x + 3 . The graph of y = fx is shown. Determine the value of f0 .

[ ] A: 1
[ ] B: -1
[x] C: 3
[ ] D: -3

Solution: To find the value of f0 , substitute x=0 into the function fx = x^2 - 4x + 3 . f0 = 0^2 - 40 + 3 f0 = 0 - 0 + 3 f0 = 3 . Therefore, the value of f0 is 3. [M1] [A1]

Q7. Consider the function fx = -2x + 6 . The graph of y = fx is shown below. Determine the value of the y -intercept.

[x] A: 6
[ ] B: -3
[ ] C: 3
[ ] D: -6

Solution: The y -intercept of a function is the value of y when x=0 . For the function fx = -2x + 6 , we substitute x=0 into the equation: f0 = -20 + 6 f0 = 0 + 6 f0 = 6 Therefore, the y -intercept is 6. [M1] [A1]

Q8. Consider the function fx = x^2 - 4x + 3. The graph of y = fx is shown below. The function intersects the x-axis at two points. Find the coordinates of the point with the smaller x-value.

[x] A: 1, 0
[ ] B: 3, 0
[ ] C: 0, 3
[ ] D: 0, 1

Solution: To find the x-intercepts of the function fx = x^2 - 4x + 3, we set fx = 0. This gives us the quadratic equation x^2 - 4x + 3 = 0. We can factor this equation as x-1x-3 = 0. The solutions are x=1 and x=3. Therefore, the x-intercepts are at 1, 0 and 3, 0. The point with the smaller x-value is 1, 0. [M1] [A1]